

A Bayesian Coalescent Model of the DNA Barcode Gap

Jarrett D. Phillips^{1,2*} (ORCID: 0000-0001-8390-386X)

Nicolas Hubert³ (ORCID: 0000-0001-9248-3377)

Robert H. Hanner² (ORCID: 0000-0003-0703-1646)

¹*School of Computer Science, University of Guelph, Guelph, ON., Canada, N1G2W1*

²*Department of Integrative Biology, University of Guelph, Guelph, ON., Canada, N1G2W1*

³*Université de Montpellier, Place Eugène Bataillon, 34095 Montpellier cedex 05, France*

***Corresponding Author:** Jarrett D. Phillips¹

Email Address: jphill01@uoguelph.ca

Running Title: Bayesian inference for DNA barcode gap coalescent estimation

Abstract

A simple statistical model of the DNA barcode gap is outlined. Specifically, accuracy of recently introduced nonparametric metrics, inspired by coalescent theory, to characterize the extent of proportional overlap/separation in maximum and minimum pairwise genetic distances within and among species, respectively, is explored in both frequentist and Bayesian contexts. The empirical cumulative distribution function (ECDF) is utilized to estimate probabilities associated with positively skewed extreme tail distance distribution quantiles bounded on the closed unit interval $[0, 1]$ based on a straightforward binomial count of overlapping specimen records. Statistical moments, notably, the mean and variance, are derived for edge cases of no overlap/complete separation and full overlap/no separation in genomic sequence variation, and are shown to be quite tight as sample sizes increase. Further, a new way to visualize distances and the DNA barcode gap estimators via accumulated probabilities appears revealing.

Using R and the probabilistic programming language, Stan, the proposed maximum likelihood estimators and Bayesian model are demonstrated on cytochrome *b* (CYTB) gene sequences from two *Agabus* diving beetle species, *A. bipusulatus* and *A. nevadensis*, exhibiting limits in the extent of representative taxonomic sampling ($N = 701$ and $N = 2$ individuals, respectively). Large-sample theory and MCMC simulations clearly expose problems, showing much uncertainty in parameter estimates, particularly under the former paradigm, and when specimen sample sizes for target species are small. Findings herein highlight the promise of the Bayesian approach using a conjugate beta prior for reliable posterior estimation over classical inference when available data are sparse. Obtained results can help shed light on foundational and applied research questions concerning DNA-based specimen identification and species delineation for studies in evolutionary biology and ecology, as well as biodiversity conservation, forensics, management and restoration, of wide-ranging taxa.

Keywords: Bayesian/frequentist inference, DNA barcoding, Hamiltonian Monte Carlo (HMC), intraspecific genetic distance, interspecific genetic distance, multispecies coalescent, specimen identification, species discovery

1 Introduction

The routine use of DNA sequences to support broad evolutionary hypotheses and questions concerning central demographic processes, like gene flow and speciation, is now ubiquitous. Since the late 1980s, mitochondrial DNA (mtDNA) has been employed to produce a distinctive and measurable signature of genetic polymorphism that readily elucidates phylogeographic patterns, such as colonization and extinction, in diverse and spatially-distributed taxonomic lineages, such as birds, fishes, insects, and arachnids, among other extensively studied groups (Avice et al., 1987). The application of genomic data to applied fields, like biodiversity forensics, conservation, management, and restoration, for the molecular identification of unknown organismal samples, came later (*e.g.*, Forensically Informative Nucleotide Sequencing (FINS); Bartlett and Davidson (1992)). Since its inception over 20 years ago, DNA barcoding (Hebert et al., 2003a) built significantly on earlier work and has emerged as a robust method of specimen identification and species discovery across myriad multicellular eukaryotes. This endeavour has dramatically expedited millions of collected specimens to be sequenced at easily obtained short, standardized gene regions like the cytochrome *c* oxidase subunit I (5'-COI) mitochondrial locus for animals (Hebert et al., 2003b) in the largest taxon-oriented initiative to date. However, despite far surpassing the depth and breadth of traditional morphospecies taxonomy at 10 times the speed, and for a fraction of the cost, DNA barcoding is not without its controversies.

The success of the single-locus approach, particularly for regulatory applications, depends crucially on two important factors: (1) the availability of high-quality specimen records found in public reference sequence databases such as the Barcode of Life Data Systems (BOLD; <http://www.barcodinglife.org>) (Ratnasingham and Hebert, 2007) and GenBank (<https://www.ncbi.nlm.nih.gov/genbank/>), and (2) the establishment of a DNA barcode gap — the notion that the maximum genetic distance observed within species is much smaller than the minimum degree of marker variation found among species (Meyer and Paulay, 2005; Meier et al., 2008). Early research has demonstrated that the presence of a DNA

barcode gap hinges strongly on extant levels of species haplotype diversity gauged from comprehensive specimen sampling at wide geographic and ecological scales (Bergsten et al., 2012; Čandek and Kuntner, 2015). Notwithstanding, many taxonomic groups lack adequate separation in their pairwise intraspecific and interspecific genetic distances due to varying rates of evolution in both genes and taxa (Pentinsaari et al., 2016). Furthermore, it has been well-demonstrated that the presence of a DNA barcode gap becomes less certain with increasing spatial scale of sampling, since interspecific distances increase due to population structuring under limited dispersal through isolation-by-distance, while intraspecific distances shrink as more closely-related species are sampled (Phillips et al., 2022). This can pose problems in cases of rare singleton species, endemics, or monotypic taxa, for instance (Ahrens et al., 2016). As a result, rapid matching of unknown query samples to expertly-validated references can be compromised, leading to cases of false positives (taxon oversplitting) and false negatives (excessive lumping of taxa) due to incomplete lineage sorting, interspecies hybridization, genome introgression, species synonymy, cryptic diversity, and specimen misidentifications (Hubert and Hanner, 2015; Phillips et al., 2022). Another culprit includes PCR/sequencing error caused by incorrect nucleotide basecalling, chimeras/heteroplasmies, sample contamination, indels (insertion/deletions), NUMT (nuclear-mitochondrial insert) pseudogene amplification, or *Wolbachia* endosymbiont infection, among other explanations (Phillips et al., 2023). Each of these can artificially magnify levels of genetic diversity in taxon populations, rendering techniques like DNA barcoding unreliable for certain groups like marine invertebrates, which are both undersampled and possess slow rates of molecular evolution (Bucklin et al., 2011).

Recent work has argued that DNA barcoding, in its current form, is lacking in statistical rigor (Phillips et al., 2022). Most publications rely strongly on heuristic distance-based thresholds calculated from convenient specimen sample sizes (usually 5-10 specimens per species, but singleton/doubletons are more prevalent). Instead, focus should be placed on optimal sampling designs using collated taxon haplotype information, coupled with project

funding and budget constraints (Phillips et al., 2015, 2019, 2020). This is necessary to accurately infer taxonomic identity and delineate boundaries on the basis of clustering genetic variation into putative groups reminiscent of species (Phillips et al., 2022). Of these studies, few report measures of uncertainty, such as standard errors (SEs) and confidence intervals (CIs), around estimates of intraspecific and interspecific variation (*e.g.*, Wiemers and Fiedler (2007)), calling into question the existence of a true species' DNA barcode gap (Čandek and Kuntner, 2015; Phillips et al., 2022). To support this idea, novel nonparametric locus- and species-specific metrics based on the multispecies coalescent (MSC) were recently outlined by Phillips et al. (2024) (see **Methods**).

The basic coalescent (Kingman, 1982a,b) encompasses a backwards continuous-time stochastic Markov process of allelic sampling among lineages under the infinite sites model of mutation within a large neutrally-evolving, panmictic, diploid species population of constant size (*i.e.*, obeying Wright-Fisher dynamics) from the present into the past towards the most recent common ancestor (MRCA). Although the coalescent sampling process plays a foundational role in evolutionary biology and population genetics theory, due to its inherent simplicity and flexibility (Rosenberg and Nordborg, 2002; Degnan and Rosenberg, 2009), the approach has been both under-utilized and under-appreciated as a key player within DNA barcoding (Collins and Cruickshank, 2014; Dowton et al., 2014; Hubert and Hanner, 2015; Phillips et al., 2022). Numerous theoretical and empirical investigations have shown that protein-coding sequence variation within non-recombining haploid loci like COI is generally subject to strong purifying selection and selective sweeps at nonsynonymous sites, rather than the occurrence of functionally silent substitutions within synonymous regions; consequently, DNA barcoding works in practice, but not in theory (Stoeckle and Thaler, 2014). However, because mitochondrial markers have considerably smaller effective population sizes (N_e) compared to nuclear regions of the genome, accumulated mutations become fixed, and therefore diagnostic of species, at a much faster rate due to genetic drift (Hubert and Hanner, 2015; Phillips et al., 2022). Previously proposed MSC algorithmic approaches (of which there

are too many to exhaustively list here, and which have mostly been applied to multilocus sequence data), generally assume either a strict or relaxed molecular clock and a model of DNA nucleotide substitution across closely-related taxa (Ho and Duchêne, 2014; Flouri et al., 2022). From here, an estimated species phylogeny may be easily constructed (*e.g.*, with or without use of a guide tree) (*e.g.*, Rannala and Yang (2003, 2017); Yang and Rannala (2010, 2014, 2017)). Even tree- and network-based taxon delimitation methods designed for single loci, such as the Generalized Mixed Yule Coalescent (GMYC) (Pons et al., 2006; Monaghan et al., 2009; Fujisawa and Barraclough, 2013), Poisson Tree Processes (PTP) (Zhang et al., 2013; Kapli et al., 2017), and statistical parsimony/haplotype networks (*e.g.*, TCS networks (Templeton et al., 1992), haplowebs (Dellicour and Flot, 2015, 2018)), which see much use in DNA barcoding initiatives, have their flaws. Optimal performance of these methods relies heavily on careful parameter selection (*e.g.*, prior specification of branching rates in ultrametric trees in the case of GMYC, or the probability of parsimony for nested clade analysis (NCA; Templeton (1998)), to name a few) (Hubert et al., 2024) within third-party software like BEAST (Bouckaert et al., 2019), among other concerns (Talavera et al., 2013; Tang et al., 2014; Fonseca et al., 2021).

Hubert et al. (2024) categorized statistical species delimitation methods according to five crucial factors necessary for success: availability, reliability, scalability, understandability, and usability. Any proposed approach should score high across all of these aspects (Hubert et al., 2024). Yet, numerous tradeoffs exist, and so current methods fall short. (See Table 1 in Hubert et al. (2024) for a comprehensive overview.) Careless parameterization of established approaches by users unfamiliar with their underlying statistical characteristics may severely bias overall results in unintended ways (Hart and Sunday, 2007). In contrast, Phillips et al.’s (2024) approach is both tree- and network-free, and does not require judicious parameter setting; all that is needed is a distance matrix. Therefore, the method is easy to use and interpret, quite memory efficient, very fast to run on large datasets, and archived freely online through GitHub (<https://github.com/jphill01/DNA-Barcode-Gap-Coalescent>). Most

importantly, the approach ranks highly across all criteria mentioned by Hubert et al. (2024).

The estimators from Phillips et al. (2024) represent a clear improvement over simple, yet arbitrary, distance heuristics such as the 2% rule noted by Hebert et al. (2003a) and the 10× rule (Hebert et al., 2004). The 2% rule asserts that DNA barcodes differing by at least 2% at sequenced genomic regions should be expected to originate from different biological species. In contrast, the 10× rule suggests that barcode sequences displaying 10 times more genetic variation among species than within taxa is suggestive of a distinct evolutionary origin. These conventions form the core of threshold-based single-locus species delimitation tools like Automatic Barcode Gap Discovery (ABGD) (Puillandre et al., 2011), Assemble Species by Automatic Partitioning (ASAP) (Puillandre et al., 2021), and the Barcode Index Number (BIN) framework (Ratnasingham and Hebert, 2013). However, the lack of adoption of an explicit and universally agreed-upon species concept in DNA barcoding studies is missing. The ideal concept is one that readily governs lineage formation and proliferation necessary to establish rigorous taxon definitions for successful delimitation of hypothesized and heuristic evolutionary units using DNA barcode identification criteria (de Queiroz, 2007; Pante et al., 2015; Rannala, 2015; Wells et al., 2022). Such a concept must be proposed in conjunction with secondary lines of evidence (*e.g.*, morphology, ecology, geography, ontogeny, and behaviour) promised by an integrative framework (Dayrat, 2005). This is the case, for instance, with species existing in allopatry/sympatry, where establishment of reproductive isolation on the basis of fixed diagnostic differences using the Biological Species Concept (BSC) (Mayr, 1942), is challenging (Fujita et al., 2012). Notably, Fujita et al. (2012) remarks that the MSC applied to the taxon delimitation problem harmonizes many species concepts since it identifies independent evolutionary trajectories under the umbrella of the General Lineage Concept. This makes the MSC a fitting tool to model speciation and other modes of diversification (Collins and Cruickshank, 2014). In addition, the reliance on visualization approaches, such as frequency histograms, dotplots, and quadrant plots, to expose DNA barcoding’s limitations, has also been criticized since they can fog real species signal relative

to noise (Collins and Cruickshank, 2013; Phillips et al., 2022). Up until the work of Phillips et al. (2024), the majority of studies (*e.g.*, Young et al. (2021)) have treated the DNA barcode gap as a binary response. Given poor sampling depth for most taxa, a Yes/No dichotomy is inherently flawed because it can falsely imply a DNA barcode gap is present for a taxon of interest when in fact no such separation in genetic distances exists. On the other hand, in the case of sequence overlap, while the presence a single specimen record is a necessary requirement to nullify the existence of a DNA barcode gap in light of available genetic data, it is not a sufficient condition, since said taxon record could represent a species misidentification or other error. Thus, the MSC serves as an ideal model to settle ongoing debates and issues regarding DNA barcoding’s use as both a molecular identification and delineation tool. The recently introduced DNA barcode gap metrics go some way into further operationalizing this task.

Phillips et al.’s (2024) proposed statistics quantify the extent of asymmetric directionality of proportional distance distribution overlap/separation for species within well-sampled taxonomic genera based on a straightforward distance count, in a similar vein to established measures of statistical similarity such as the Kullback-Leibler (KL) divergence (Kullback and Leibler, 1951) and other related statistics. The metrics can be employed in a variety of ways, including to validate performance of marker genes for specimen identification to the species level (using nonparametric bootstrapping, as in Phillips et al. (2024)). The estimators can further be utilized to assess whether computed values are consistent with fundamental population genetic-level parameters like N_e , mutation rates (μ), migration rates (m), and divergence times (τ), as well as derived ones, such as the population mutation rate $\theta = 4N_e\mu$, and the fixation index, F_{ST} . This is especially true for species under study within a statistical phylogeographic setting (Knowles and Maddison, 2002; Knowles, 2004, 2009; Mather et al., 2019). Early on, species discovery using DNA barcoding was presumed to only work for reciprocally monophyletic groups, and thus concerned itself with terminal branches of generated phylogenies rather than more basal lineages occurring deeper in hypothesized

species trees (Fujita et al., 2012; Mutanen et al., 2016). Furthermore, the occurrence of short branches and large N_e within resolved phylogenies increases the likelihood of deep coalescence (Degnan and Rosenberg, 2009), clouding species delimitations, which often fail or are uncertain in broad parameter space (Carstens et al., 2013; Hickerson et al., 2006; Rannala, 2015). As DNA barcoding is a single-locus approach, it is problematic for evolutionarily young taxa. Incomplete lineage sorting within gene genealogies, leading to the retention of ancestral polymorphisms within populations, is a common phenomenon due to the ongoing stochastic dynamic of mutation generating population variation, and genetic drift driving gene variants to fixation (Degnan and Rosenberg, 2009; Rannala, 2015). The most promising way forward in assessing the validity of Phillips et al.’s (2024) method seems to be through the use of software such as BPP (Bayesian Phylogenetics and Phylogeography), which permits efficient full Bayesian simulations under various MSC models (*e.g.*, MSC-I (MSC with introgression; Flouri et al. (2020)) or MSC-M (MSC with migration; Flouri et al. (2023)), among others) using Markov Chain Monte Carlo (MCMC) for tree parameter estimation (via the A00 option, for instance) (Flouri et al., 2018), or PHRAPL (Phylogeographic Inference using Approximate Likelihoods) (Jackson et al., 2017a,b), which employs tractable phylogenetic likelihood calculations via the genealogical divergence index (gdi). Other hypothesis-driven approaches to species delimitation model selection, like Bayes Factors (BFs) (Leaché et al., 2014) and Approximate Bayesian Computation (Camargo et al., 2012), are also possible. Rigorous statistical methodology has been historically integrated into DNA barcoding previously (Matz and Nielsen, 2005; Nielsen and Matz, 2006), though strictly for the purpose of specimen identification. Phillips et al.’s (2024) is unique as it also greatly extends usage to delimit species, while still being grounded strongly in evolutionary theory.

While introduction of the new metrics is a step in the right direction, what appears to be missing is a rigorous statistical treatment of the DNA barcode gap. This includes an unbiased way to compute the statistical accuracy of Phillips et al.’s (2024) estimators arising through problems inherent in frequentist maximum likelihood estimation for probability distributions

having bounded positive support on the closed unit interval $[0, 1]$. To this end, here, a Bayesian model of the DNA barcode gap coalescent is introduced to rectify such issues. The model allows accurate estimation of posterior means, posterior standard deviations (SDs), posterior quantiles (*e.g.*, median) and credible intervals (CrIs) for the statistics given datasets of intraspecific and interspecific distances for species of interest.

2 Methods

2.1 DNA Barcode Gap Metrics

The novel nonparametric maximum likelihood estimators (MLEs) of proportional overlap/separation between intraspecific and interspecific distance distributions for a given species (x) to aid assessment of the DNA barcode gap are as follows:

$$p_x = \frac{\#\{d_{ij} \geq a\}}{\#\{d_{ij}\}} \quad (1)$$

$$q_x = \frac{\#\{d_{XY} \leq b\}}{\#\{d_{XY}\}} \quad (2)$$

$$p'_x = \frac{\#\{d_{ij} \geq a'\}}{\#\{d_{ij}\}} \quad (3)$$

$$q'_x = \frac{\#\{d'_{XY} \leq b\}}{\#\{d'_{XY}\}} \quad (4)$$

where d_{ij} are distances within species, d_{XY} are distances among species for an *entire* genus of concern, and d'_{XY} are combined interspecific distances for a target species and its closest neighbouring species. The notation $\#$ reflects a count. Distances form a continuous distribution and are easily computed from a model of DNA sequence evolution, such as uncorrected or corrected p-distances (Jukes and Cantor, 1969; Kimura, 1980) using, for example, the `dist.dna()` function available in the `ape` R package (Paradis et al., 2004). Quantities a , a' , and b correspond to $\min(d_{XY})$, $\min(d'_{XY})$, and $\max(d_{ij})$, the minimum

246 interspecific distance, the minimum combined interspecific distance, and the maximum
 247 intraspecific distance, respectively (**Figure 1**). Note, distances are constrained to the interval
 248 $[0, 1]$, whereas the metrics are defined only on the interval $[a/a', b]$.

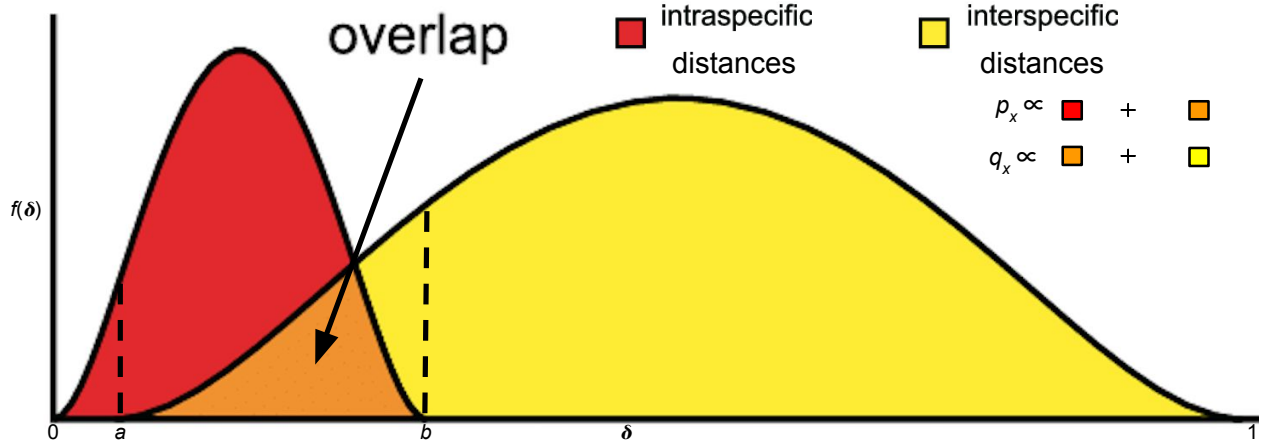


Figure 1: Modified depiction from Meyer and Paulay (2005) and Phillips et al. (2024) of the overlap/separation of intraspecific and interspecific distances (δ) for calculation of the DNA barcode gap metrics (p_x and q_x) for a hypothetical species x . The minimum interspecific distance is denoted by a and the maximum intraspecific distance is indicated by b . The quantity $f(\delta)$ is akin to a kernel density estimate (KDE) of the probability density function of distances. A similar visualization can be displayed for p'_x and q'_x based on intraspecific and combined interspecific distances within the interval $[a', b]$.

249 Values of the estimators obtained from Equations (1)-(4) close to or equal to zero give
 250 evidence for separation between intraspecific and interspecific distance distributions; that is,
 251 values suggest the presence of a DNA barcode gap for a target species. Conversely, values
 252 near or equal to one give evidence for distribution overlap; that is, values likely indicate the
 253 absence of a DNA barcode gap. Furthermore, calculation of the DNA barcode gap estimators
 254 is convenient as they implicitly account for total distribution area within $[a/a', b]$ (including
 255 the overlap region) (**Figure 1**). It should be noted however that there is a loss of information
 256 in computing estimates of Phillips et al.'s (2024) metrics since two different sets of (p_x, q_x)
 257 and (p'_x, q'_x) values cause equal but distinct underlying distance distributions of $f_{\text{intra}}(\delta)$ and
 258 $f_{\text{inter}}(\delta)$ (**Figure 1**). In other words, under certain circumstances, the metrics cannot be
 259 determined uniquely, and so are non-identifiable in these cases.

Hence, Equations (1)-(4) are simply empirical partial means of distances falling at and below,
 or at and exceeding, given distribution thresholds. Notice further that a/a' , and b are also
 the first and n th order statistics, $X_{(1)}$ and $X_{(n)}$, respectively, with $a/a' < b$, which have been
 pointed out by Phillips et al. (2022) as important for developing a statistical theory to test
 for the existence of the DNA barcode gap. However, the underlying sampling distributions of
 $\eta = a - b$ and $\eta' = a' - b$ are not obvious (they involve the evaluation of convolutions). This
 caveat renders a formal hypothesis test cumbersome to derive since said distributions are
 not expected to be symmetric. The standard bootstrap resampling procedure (Efron, 1979)
 can also fail when the estimator of interest is not well-behaved, necessitating extensions
 like the m -out-of- n bootstrap discussed briefly in Phillips et al. (2022). This is similar
 to calculating bootstrap support values to assess how well tree nodes justify present data
 (Efron et al., 1996). However, as more data are collected, support values, along with their
 associated SEs, will necessarily converge to unity and zero, respectively. Phillips et al. (2024)
 made an argument for the resampling of alignment sites since DNA barcodes are short in
 length and sites contributing to distance distribution overlap may not be included within
 generated bootstrap resamples; yet, no separation could still be evident. Equations (1)-(4)
 can also be expressed in terms of empirical cumulative distribution functions (ECDFs) (see
 next section). A clear connection between the ECDF and the nonparametric MLE was
 noted by Efron and Tibshirani (1993) as well as others. However, calculated values are *not*
 independent and identically distributed (IID) because estimated standard errors (SEs) will
 depend on both the number of species sampled within the genus under study, as well as the
 number of specimens sampled within a target species. To tease this out, Phillips et al. (2024)
 suggests plotting estimator values against their estimated SEs, along with a simple random
 downsampling scheme. In the case of two species comprising a focal genus, one well sampled
 and the other poorly sampled, values of the metrics close to zero for the sufficiently sampled
 species will likely possess larger SEs following downsizing to match the number of poorly
 sampled specimens (Phillips et al., 2024). While this scheme appears promising, challenges

287 arise in uncertainty quantification as discussed later.

288 2.2 Relationship to (Multispecies) Coalescent Theory

289 The approach of Phillips et al. (2024) differs markedly from the traditional definition of
290 the DNA barcoding gap laid out by Meyer and Paulay (2005) and Meier et al. (2008) in that
291 the proposed metrics incorporate interspecific distances which *include* the target species of
292 interest. Furthermore, if a focal species is found to have multiple nearest neighbours, then the
293 species possessing the smallest average distance is used (Phillips et al., 2024). These schemes
294 more accurately account for species' and coalescent histories inferred from contemporaneous
295 samples of DNA sequences leading to instances of barcode sequence sharing, such as
296 interspecific hybridization/introgression events (Phillips et al., 2024). For example, within
297 Equations (3) and (4), the degree of distance distribution overlap/separation between a
298 target taxon and its nearest neighbouring species, gauged from magnitudes of p'_x and q'_x ,
299 is directly proportional to the amount of time in which the two lineages diverged from the
300 MRCA (Phillips et al., 2024). More specifically, $a \propto t_{XY, \min}$ and $b \propto t_{ij, \max}$, the minimum
301 and maximum (unobserved) divergence time (in units of N_e generations), respectively, for
302 two different individuals (i, j) and species (X, Y) within a well-inventoried genus (Phillips
303 et al., 2024). Figure 1 in Phillips et al. (2024) depicts a phylogeny for two hypothetical
304 species, A and B , showing paraphyly based on the sequencing of two mitochondrial loci.
305 The (unobserved) time the two species formed, t_{AB} , is intermediate between $t_{XY, \min}$ and
306 $t_{ij, \max}$, but theoretically, its maximum ($t_{AB, \max}$) may occur more distantly in the past closer
307 to the root comprising the MRCA for all sampled taxa (Phillips et al., 2024). From a
308 coalescence perspective, p_x is the probability two lineages within species x coalesce after
309 $t_{XY, \min}$, whereas q_x represents the probability a lineage within species x coalesces with a
310 lineage in species k (within a sample) after $t_{XY, \min}$, but before $t_{ij, \max}$. This approach is
311 similar to that of De Sanctis et al. (2022), who devised a coalescent model of taxonomic
312 binning assignment accuracy given query and reference DNA sequences for two species found

in a well-curated genomic database. However, the approach herein is inherently simpler, easily extensible, and more intuitive, since it does not make any parametric assumptions regarding mutations along branch lengths, coalescence times, *etc.* and can be applied to more than two taxa in tandem. Thus, the quantities can be used as a criterion to assess the failure of DNA barcoding in recently radiated taxonomic groups, for example, among other plausible biological explanations mentioned earlier.

2.3 The Model

2.3.1 Relating the DNA Barcode Gap Metrics to the Cumulative Distribution Function

Before delving into the derivation of the proposed DNA barcode gap metrics, review of some fundamental statistical theory is necessary. For a given random variable X from a specified continuous probability distribution function (PDF), f , its cumulative distribution function (CDF), F , is defined by

$$F_X(t) = \int_{-\infty}^t f(t) dt = \mathbb{P}(X \leq t) = 1 - \mathbb{P}(X > t). \quad (5)$$

Rearranging Equation (5) gives

$$\mathbb{P}(X > t) = 1 - F_X(t) = 1 - \mathbb{P}(X \leq t), \quad (6)$$

from which it follows that

$$\mathbb{P}(X \geq t) = 1 - F_X(t) + \mathbb{P}(X = t). \quad (7)$$

328 The CDF has many desirable statistical properties, such as boundedness, continuity, and
 329 monotonicity. Consult any graduate-level text on mathematical statistics or probability
 330 theory for reference.

331 Equations (1)-(4) can thus be written in terms of ECDFs via Equations (5) and (7) as
 332 follows, since the true underlying F s, are unknown *a priori*, and therefore must be estimated
 333 using available sequence distance data:

$$\begin{aligned} p_x &= \mathbb{P}(d_{ij} \geq a) \\ &= 1 - \hat{F}_{d_{ij}}(a) + \mathbb{P}(d_{ij} = a) \end{aligned} \quad (8)$$

$$\begin{aligned} q_x &= \mathbb{P}(d_{XY} \leq b) \\ &= \hat{F}_{d_{XY}}(b) \end{aligned} \quad (9)$$

$$\begin{aligned} p'_x &= \mathbb{P}(d_{ij} \geq a') \\ &= 1 - \hat{F}_{d_{ij}}(a') + \mathbb{P}(d_{ij} = a') \end{aligned} \quad (10)$$

$$\begin{aligned} q'_x &= \mathbb{P}(d'_{XY} \leq b) \\ &= \hat{F}_{d'_{XY}}(b). \end{aligned} \quad (11)$$

334 Clearly, $\hat{F}_{d_{ij}}(b) = \hat{F}_{d_{XY}}(1) = 1$ and $\hat{F}_{d_{XY}}(a) = \hat{F}_{d_{ij}}(0) = 0$. Given n increasing-ordered data
 335 points, the (discrete) ECDF, $\hat{F}_n(t) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{[x_i \leq t]}$, comprises an increasing step function
 336 having jump discontinuities of size $\frac{1}{n}$ at each sample observation (x_i) , excluding ties (or
 337 steps of weight $\frac{i}{n}$ with duplicate observations), where $\mathbb{1}(x)$ is the indicator function. Note
 338 that $1 - \hat{F}_n(t)$ is the complementary ECDF (CECDF), which is a declining function. Further,
 339 $\mathbb{P}(X = t)$ is the probability that X (a distance) takes on the value t . The ECDF/CECDF has
 340 good asymptotic behaviour in the limit of sample size. Equations (8)-(11) clearly demonstrate
 341 the asymmetric directionality of the proposed metrics. Plots of the ECDFs/CECDFs for
 342 nonzero intraspecific, interspecific, and combined interspecific distances, superimposed with

corresponding values of a/a' and b , provide a new, useful way to visualize the DNA barcode gap and Phillips et al.'s (2024) statistics. Firstly, there is no need to tune the output (*e.g.*, number of bins and bin width in the case of histograms (Phillips et al., 2022)). Secondly, probabilities reflecting areas under the distance curves depicted in **Figure 1** can be directly read off the ECDF/CECDF curves. Values of the metrics can also be determined from direct inspection of the ECDF/CECDF plot by determining the corresponding y -value (cumulative probability) for a given value of x (distance).

2.3.2 Integrating Frequentist and Bayesian Inference

A major criticism of large sample (frequentist) theory is that it relies on asymptotic properties of the MLE (whose population parameter is assumed to be a fixed but unknown quantity), such as estimator normality and consistency as the sample size approaches infinity. This problem is especially pronounced in the case of binomial proportions (Newcombe, 1998). The estimated Wald SE of the sample proportion, is given by $\widehat{SE}[\hat{p}] = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$, where $\hat{p} = \frac{Y}{n}$ is the MLE, Y is the total number of successes ($Y = \sum_{i=1}^n y_i$), and n is the total number of trials (*i.e.*, sample size). However, the above formula for the standard error is problematic for several reasons. First, it is a Normal approximation which makes use of the central limit theorem (CLT); thus, large sample sizes are required for reliable estimation. When few observations are available, SEs will be large and inaccurate, leading to low statistical power to detect a true DNA barcode gap when one actually exists. Further, resulting interval estimates could span values less than zero or greater than one, or have zero width, which is practically meaningless. Second, when proportions are exactly equal to zero or one, resulting SEs will be exactly zero, rendering $\widehat{SE}[\hat{p}]$ given above completely useless. In the context of the proposed DNA barcode gap metrics, values obtained at the boundaries of their support are often encountered. Therefore, reliable calculation of SEs is not feasible. Given the importance of sufficient sampling of species genetic diversity for DNA barcoding initiatives, a different statistical estimation approach is necessary.

Bayesian inference offers a natural path forward in this regard since it allows for straightforward specification of prior beliefs concerning unknown model parameters and permits the seamless propagation of uncertainty, when data are lacking and sample sizes are small, through integration with the likelihood function associated with true generating processes. The posterior distribution ($\pi(\theta|Y)$) is given by Bayes' theorem up to a constant of proportionality $\pi(\theta|Y) \propto \pi(Y|\theta)\pi(\theta)$, where θ are unobserved parameters, Y are known data, $\pi(Y|\theta)$ is the likelihood, and $\pi(\theta)$ is the prior. As a consequence, because parameters are treated as random variables, Bayesian models are much more flexible and generally more easily interpretable compared to frequentist approaches. Under the Bayesian paradigm, entire posterior distributions, along with their statistical summaries (*e.g.*, CrIs) are outputted, rather than just long run behaviour reflected in sampling distributions, p-values, and CIs as in the frequentist case, thus allowing direct probability statements to be made.

2.3.3 Relating the DNA Barcode Gap Metrics to Mixture Models

Essentially, from a statistical perspective, the goal herein is to nonparametrically estimate probabilities corresponding to extreme tail quantiles for positive highly skewed distributions on the unit interval (or any closed subinterval thereof). Here, it is sought to numerically approximate the extent of proportional overlap/separation of intraspecific and interspecific distance distributions within the subinterval $[a/a', b]$. This is a challenging computational problem within the current study as detailed in subsequent sections. The usual approach employs kernel density estimation (KDE), along with numerical or Monte Carlo integration of complicated PDFs, and invocation of extreme value theory (EVT) to calculate overlap (such as that for species with unusually large distances suggestive of cryptic species diversity, or deep sequence divergence due to long periods of genetic isolation); however, this requires careful selection of the bandwidth parameter, which would need to be identical for both distributions, among other considerations. This becomes problematic when fitting finite mixture models using MCMC, where nonidentifiability is rampant due to the occurrence

of label switching, making clusters hard to discern across sampling iterations, and model selection challenging due to lack of exchangeability among observations (Stephens, 2000). For DNA barcode gap estimation, this would correspond to a two-component mixture (one for intraspecific distance comparisons, and the other for interspecific comparisons), with one or more distribution modes or curve intersection points between components, and the presence of zero distance inflation. This makes parameter estimation difficult using methods like the Expectation-Maximization (EM) algorithm (Dempster et al., 1977) as the algorithm may become stuck in suboptimal regions of the parameter search space and prematurely converge to local optima.

The abovementioned statistical issues seems wholly plausible, especially given that the genus-level distribution of genetic distances is a mixture of closely-related (*i.e.*, nearest neighbour) and more distant species. Examining **Figure 1**, depending on divergence times and other elements, intraspecific and interspecific distributions would be expected to comprise greater density in the upper (right) tails (which make up more distantly-related taxa relative to a target species) compared to the lower (left) tails (which contain only the closest conspecifics) since there are more combinations of distance comparisons possible. A key problem affecting reliable inference of the DNA barcode gap metrics is undersampling of genus-level and/or interspecific variation. An immediate, but non-trivial solution appears to be a move toward next-generation DNA barcoding incorporating multiple loci (Collins and Cruickshank, 2014; Dowton et al., 2014). In theory, this would facilitate the detection of missing operational taxonomic units (OTUs), both extant and extinct, when attempting to better characterize a genus. In this context, Phillips et al.’s (2024) statistics could be informative in this regard for studies of human phylogeography and ancestry composition where historical admixture between sympatric populations was commonplace (*e.g.*, modern human/Denisova interbreeding). Here, for simplicity, an alternate route is taken to avoid these obstacles.

2.3.4 Bayesian Model Setup

Counts, y , of overlapping distances (as expressed in the numerator of Equations (1)-(4)) are treated as binomially distributed; that is, $\pi(y|\theta) = \binom{n}{y}\theta^y(1-\theta)^{n-y}$, where $\binom{n}{y}$ is the binomial coefficient expressing the number of ways to choose y elements from n total objects. Total counts have expectation $\mathbb{E}[Y|\theta] = n\theta$, and variance $\mathbb{V}[Y|\theta] = n\theta(1-\theta)$, where $n = \{N, C\}$ are total count vectors of intraspecific and combined interspecific distances, respectively, for a target species along with its nearest neighbour species, and $n = M$ is a total count vector for all interspecific species comparisons. This follows from the fact that the ECDF/CECDF is binomially distributed. The quantity thus being estimated is the parameter vector $\underline{\theta} = \{p_x, q_x, p'_x, q'_x\}$. The metrics encompassing $\underline{\theta}$ are presumed to follow a Beta(α, β) distribution, with real shape parameters α and β , which is a natural choice of prior on probabilities. That is, $\pi(\theta|\alpha, \beta) = \frac{1}{B(\alpha, \beta)}\theta^{\alpha-1}(1-\theta)^{\beta-1}$, where $B(\alpha, \beta)$ is the beta function evaluated at α and β . The beta distribution has a prior mean of $\mathbb{E}[\theta|\alpha, \beta] = \frac{\alpha}{\alpha+\beta}$ and a prior variance equal to $\mathbb{V}[\theta|\alpha, \beta] = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$. In the case where $\alpha = \beta$, all generated Beta(α, β) distributions will possess the same prior expectation, whereas the prior variance will shrink as both α and β increase. Such a scheme is quite convenient since the beta distribution is conjugate to the binomial distribution. Thus, the posterior distribution is also beta distributed, specifically, $\pi(\theta|y, \alpha, \beta) = \theta^{\alpha+y-1}(1-\theta)^{\beta+n-y-1}$ (*i.e.*, Beta($\alpha + Y, \beta + n - Y$)), having expectation $\mathbb{E}[\theta|Y, \alpha, \beta] = \frac{\alpha+Y}{\alpha+\beta+n}$ and variance $\mathbb{V}[\theta|Y, \alpha, \beta] = \frac{(\alpha+Y)(\beta+n-Y)}{(\alpha+\beta+n)^2(\alpha+\beta+n+1)}$. In the context of DNA barcoding, it is important that the DNA barcode gap metrics effectively differentiate between extremes of no overlap/complete separation and full overlap/no separation, corresponding to values of the metrics equal to 0 and 1 (equivalent to total distance counts of 0 and n), respectively. These extremes yield a posterior expectation of $\mathbb{E}[\theta|Y = 0, \alpha, \beta] = \frac{\alpha}{\alpha+\beta+n}$ and a posterior variance of $\mathbb{V}[\theta|Y = 0, \alpha, \beta] = \frac{\alpha(\beta+n)}{(\alpha+\beta+n)^2(\alpha+\beta+n+1)}$ and $\mathbb{E}[\theta|Y = n, \alpha, \beta] = \frac{\alpha+n}{\alpha+\beta+n}$ and $\mathbb{V}[\theta|Y = n, \alpha, \beta] = \frac{(\alpha+n)\beta}{(\alpha+\beta+n)^2(\alpha+\beta+n+1)}$. Note, the posterior variances are equivalent at these thresholds for all $\alpha = \beta$.

Parameters were given an uninformative Beta(1, 1) prior, which is identical to a standard uniform (Uniform(0, 1)) prior since it places equal probability on all parameter values within its support. This distribution has an expected value of $\mu = \frac{1}{2} = 0.500$ and a variance of $\sigma^2 = \frac{1}{12} \approx 0.083$. Further, the posterior is Beta($Y+1$, $n-Y+1$), from which various moments such as the expected value $\mathbb{E}[\theta|Y] = \frac{Y+1}{n+2}$ and variance $\mathbb{V}[\theta|Y] = \frac{(Y+1)(n-Y+1)}{(n+2)^2(n+3)}$, and other quantities, can be easily calculated. Clearly, $\mathbb{E}[\theta|Y=0] = \frac{1}{n+2}$ and $\mathbb{V}[\theta|Y=0] = \frac{n+1}{(n+2)^2(n+3)}$, and $\mathbb{E}[\theta|Y=n] = \frac{n+1}{n+2}$ and $\mathbb{V}[\theta|Y=n] = \frac{n+1}{(n+2)^2(n+3)}$. For example, with a single distance ($n = 1$) for a given species (wherein two specimens are compared), and where the distance is not overlapping, on average, θ will be equal to $\mu = \frac{1}{3} = 0.333$, with a variance of $\sigma^2 = \frac{2}{36} = \frac{1}{18} \approx 0.056$. In contrast, with $n = 100$ individuals and there is no overlap, the mean reduces to $\mu = \frac{1}{102} \approx 0.01$, with a variance of $\sigma^2 = \frac{101}{1071612} = 9.425 \times 10^{-5}$. When $n = 1$ and there is overlap, the mean is $\mu = \frac{2}{3} = 0.667$, having a variance of $\sigma^2 \approx 0.056$. For $n = 100$ where all distances overlap, θ has expectation $\mu = \frac{101}{102} \approx 0.990$ and variance $\sigma^2 \approx 9.425 \times 10^{-5}$. These edge case examples have good asymptotic behaviour with increasing sample size (variance tends to zero), and demonstrate that the Beta(1, 1) distribution serves as a reasonable first-pass prior for DNA barcode gap estimation using Phillips et al.'s (2024) metrics, though tighter bounds can likely be achieved. In general however, when possible, it is always advisable to incorporate prior information, even if only weak, rather than simply imposing complete ignorance in the form of a flat prior distribution. In the case of unimodal distributions, the (estimated) posterior mean often possesses the property that it readily decomposes into a convex linear combination, in the form of a weighted sum, of the (estimated) prior mean and the MLE. That is

$$\hat{\mu}_{\text{posterior}} = w\hat{\mu}_{\text{prior}} + (1-w)\hat{\mu}_{\text{MLE}}, \text{ where, for the beta distribution, } w = \frac{\alpha+\beta}{\alpha+\beta+n}.$$

Therefore, with sufficient data, $w \rightarrow 0$ as $n \rightarrow \infty$, regardless of the values of α and β , and the choice of prior distribution becomes less important since the posterior will be dominated by the likelihood. For the Beta(1, 1), $w = \frac{2}{2+n}$, with $n = 1$ giving $w = \frac{2}{3} = 0.667$. For $n = 100$, the prior weight is $w = \frac{2}{102} \approx 0.0196$. The full Bayesian model for species x is thus given by

$$\begin{aligned}
y_{\text{lwr}} &\sim \text{Binomial}(N, p_{\text{lwr}}) \\
y_{\text{upr}} &\sim \text{Binomial}(M, p_{\text{upr}}) \\
y'_{\text{lwr}} &\sim \text{Binomial}(N, p'_{\text{lwr}}) \\
y'_{\text{upr}} &\sim \text{Binomial}(C, p'_{\text{upr}}) \\
p_{\text{lwr}}, p_{\text{upr}}, p'_{\text{lwr}}, p'_{\text{upr}} &\sim \text{Beta}(1, 1)
\end{aligned} \tag{12}$$

Note that p_x , q_x , p'_x , and q'_x in Equations (1)-(4) are denoted p_{lwr} , p_{upr} , p'_{lwr} , and q'_{upr} , respectively within Equation (12) for easy distinction between MLEs and Bayesian posterior estimates. Further, y_{lwr} , y_{upr} , y'_{lwr} , and y'_{upr} are calculated from the numerators of Equations (1)-(4), respectively. The above statistical theory and derivations lay a good foundation for the remainder of this paper.

2.3.5 Computational Details

The proposed model is inherently vectorized to allow processing of multiple species datasets simultaneously. Model fitting was achieved using the Stan probabilistic programming language (Carpenter et al., 2017) framework for Hamiltonian Monte Carlo (HMC) (Neal, 2011) via the No-U-Turn Sampler (NUTS) sampling algorithm (Hoffman and Gelman, 2014) through the `rstan` R package (version 2.32.6) (Stan Development Team, 2023) in R (version 4.4.1) (R Core Team, 2024). Four Markov chains were run for 2000 iterations each in parallel across four cores with random parameter initializations. Within each chain, a total of 1000 samples was discarded as warmup (*i.e.*, burnin) to reduce dependence on starting conditions and to ensure posterior samples are reflective of the equilibrium distribution. Further, 1000 post-warmup draws were utilized per chain during the sampling phase. Because HMC/NUTS results in dependent samples that are minimally autocorrelated, chain thinning is not required. Each of these tuning parameters reflect default MCMC settings in Stan

to control both bias and variance, respectively, in the resulting draws. All analyses in the present work were carried out on a 2023 Apple MacBook Pro with M2 chip and 16 GB RAM running macOS Ventura 13.2. A random seed was set to ensure reproducibility of model results. Outputted estimates were rounded to three decimal places of precision. Posterior draws and posterior distributions of the metrics were visualized as scatterplots and KDE plots, respectively, using the `ggplot()` function within the `ggplot2` R package (version 3.5.1) (Wickham, 2016) with the default Gaussian kernel and optimal smoothness selection. To successfully run the Stan program, end users must have installed an appropriate compiler (such as GCC or Clang) which is compatible with their operating system, such as macOS.

Convergence was assessed both visually and quantitatively as follows: (1) through examining parameter traceplots, which depict the trajectory of accepted MCMC draws as a function of the number of iterations, (2) through monitoring the Gelman-Rubin potential scale reduction factor statistic ($\hat{R}$) (Gelman and Rubin, 1992; Vehtari et al., 2021), which measures the concordance of within-chain *versus* between-chain variance, and (3) through calculating the effective sample size (ESS) for each parameter, which quantifies the number of independent samples generated Markov chains are equivalent to. Mixing of chains was deemed sufficient when traceplots looked like “fuzzy caterpillars”, $\hat{R} < 1.01$, and effective sample sizes were reasonably large (Gelman et al., 2020). After sampling, a number of summary quantities were reported, including posterior means, posterior SDs, and posterior quantiles from which 95% CrIs could be computed to make probabilistic inferences concerning true population parameters. To validate the overall correctness of the proposed statistical model given by Equation (12), as a means of comparison, posterior predictive checks (PPCs) were also employed to generate binomial random variates in the form of mean counts from the posterior predictive distribution; that is $\underline{\gamma} = \{Np_x, Mq_x, Np'_x, Cq'_x\}$ to verify that the model adequately captures relevant features of the observed data.

2.3.6 Probabilistic Interpretation of the DNA Barcode Gap Metrics

The proposed Bayesian model outlined here has a straightforward interpretation (Table 1).

Table 1: Interpretation of the DNA barcode gap estimators within $[a/a', b]$

Parameter	Explanation
p_x/p_{lwr}	When p_x/p_{lwr} is close to 0 (1), it suggests that the probability of intraspecific (interspecific) distances being larger (smaller) than interspecific (intraspecific) distances is low (high) on average, while the probability of interspecific (intraspecific) distances being larger (smaller) than intraspecific (interspecific) distances is high (low) on average; that is, there is (no) evidence for a DNA barcode gap.
q_x/p_{upr}	When q_x/p_{upr} is close to 0 (1), it suggests that the probability of interspecific (intraspecific) distances being larger (smaller) than intraspecific (interspecific) distances is high (low) on average, while the probability of intraspecific (interspecific) distances being larger (smaller) than interspecific (intraspecific) distances is low (high) on average; that is, there is (no) evidence for a DNA barcode gap.
p'_x/p'_{lwr}	When p'_x/p'_{lwr} is close to 0 (1), it suggests that the probability of intraspecific (combined interspecific distances for a target species and its nearest neighbour species) distances being larger (smaller) than combined interspecific distances for a target species and its nearest neighbour species (intraspecific distances) is low (high) on average, while the probability of combined interspecific distances for a target species and its nearest neighbour species (intraspecific distances) being larger (smaller) than intraspecific distances (combined interspecific distances for a target species and its nearest neighbour species) is high (low) on average; that is, there is (no) evidence for a DNA barcode gap.
q'_x/p'_{upr}	When q'_x/p'_{upr} is close to 0 (1), it suggests that the probability of combined interspecific distances for a target species and its nearest neighbour species (intraspecific distances) being larger (smaller) than intraspecific distances (combined interspecific distances for a target species and its nearest neighbour species) is high (low) on average, while the probability of intraspecific distances (combined interspecific distances for a target species and its nearest neighbour species) being larger (smaller) than combined interspecific distances for a target species and its nearest neighbour species (intraspecific distances) is low (high) on average; that is, there is (no) evidence for a DNA barcode gap.

2.4 *Agabus* Case Study

The DNA barcode gap statistics have been shown to hold strong promise for reliable DNA barcode gap assessment when applied to the predatory diving beetles *Agabus* (Coleoptera: Dytiscidae) (Phillips et al., 2024). With a Holarctic range, despite their ease of sampling and well-established taxonomy, this group possesses few morphologically-distinct taxonomic characters that readily facilitate their assignment to the species level (Bergsten et al., 2012). Accurate taxon discrimination within *Agabus* is paramount since several species are currently classified as endangered on the International Union for Conservation of Nature (IUCN) Red List (<https://www.iucnredlist.org>) due to habitat fragmentation/loss, water pollution, and climate change. Further, the proposed metrics indicate that sister species pairs from this taxon are often difficult to distinguish on the basis of their DNA barcode sequences (Phillips et al., 2024), which may be caused by *Agabus*'s increased degree of non-monophyly at wider geospatial scales, occurrence of hybridization/introgression, or presence of synonymy (Bergsten et al., 2012). Using sequence data from three mitochondrial cytochrome markers (5'-COI, 3'-COI, and cytochrome *b* (CYTB)) obtained from BOLD and GenBank, results from Phillips et al. (2024) highlight that DNA barcoding has been a one-sided argument. Phillips et al.'s (2024) findings point to the need to balance both the sufficient collection of specimens, as well as the extensive sampling of species. Not surprisingly, DNA barcode libraries for most taxonomic groups are biased toward the latter effort (Phillips et al., 2015, 2019, 2022).

The *Agabus* CYTB dataset analyzed by Phillips et al. (2024) is revisited herein for the species *A. bipustulatus* and *A. nevadensis*, since these taxa were the sole representatives for this locus, with the most and the least specimen records, respectively ($N = 701$ and $N = 2$) across all three assessed molecular markers. Briefly, using the R package **MACER** (Young et al., 2021), DNA sequences were downloaded from GenBank and BOLD and processed using a stringent workflow to obtain a 343-bp high-quality FASTA alignment representing 46 unique haplotypes. The resulting alignment contained no stop codons,

ambiguous nucleotides, or other problematic artifacts that bias genetic diversity estimates. Further, both species- and genus-level distance outliers were excluded based on the interquartile range (IQR) (Young et al., 2021). Genetic distances were calculated using uncorrected p-distances. *A. bipustulatus* comprised 45 total haplotypes, whereas *A. nevadensis* possessed only one haplotype. Note, DNA barcode gap estimation is only possible for species having at least two specimen records because only one distance can be calculated. This dataset is a prime illustrative example highlighting the issue of inadequate taxon sampling, which arises frequently in large-scale phylogenetic and phylogeographic studies, in several respects. First, from a statistical viewpoint, sample sizes reflect extremes in reliable parameter estimation. Second, from a DNA barcoding perspective, *Agabus* currently comprises about 200 extant Linnean species according to the Global Biodiversity Information Facility (GBIF) (<https://www.gbif.org>); yet, due to the level of convenience sampling inherent in taxonomic collection efforts for this genus, adequate representation of species and genetic diversity is far from complete. Indeed, herein, species coverage for CYTB is only around 1.000%. Not surprisingly, CYTB performed poorly as a DNA barcode locus for *Agabus* despite having the highest specimen representativeness for a given species ($N = 701$, *A. bipustulatus*) and displaying good separation of genetic sequences among all three mitochondrial genes examined in Phillips et al.'s (2024) study; yet no barcode gap was observed since all metrics were found to be equal to 1.000.

3 Results and Discussion

ECDF/CECDF plots were generated using the base R function `ecdf()`, along with `ggplot()`, and are displayed in **Figure 2**. Interpretation is simple: in the case of *A. bipustulatus*, for both p_x and p'_x , all distances are greater than $a = 0$, with 36.900% of distances equal to a/a' . Conversely, both q_x and q'_x equal one because all interspecific distances are less than $b \approx 0.026$; for *A. nevadensis*, both q_x and q'_x are approximately equal

to 0.010 since only about 1% of distances fall below $b = 0$. In the case of p_x and p'_x , both statistics equal 1.000 since all distances were found to be zero.

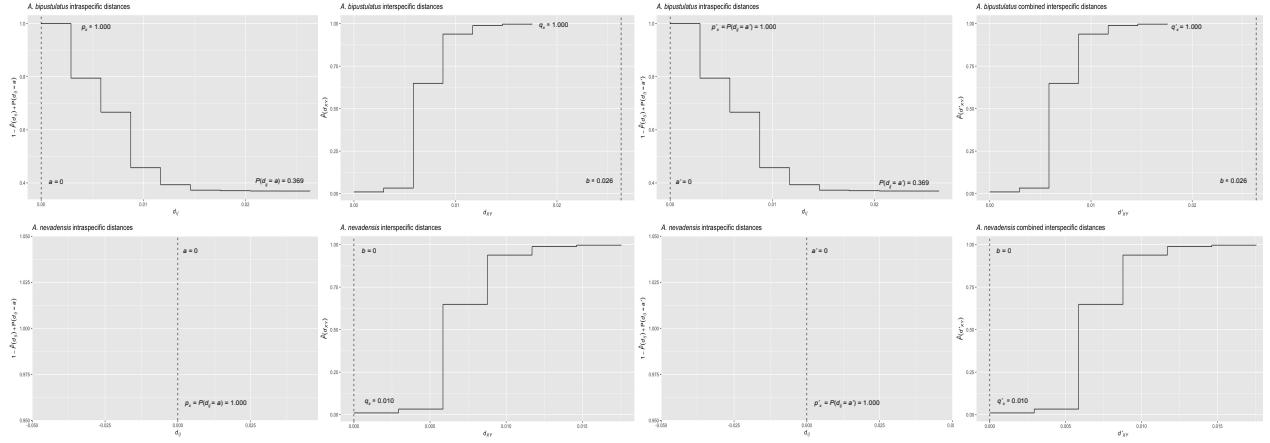


Figure 2: ECDF/CECDF plots of the DNA barcode gap metrics for *A. bipustulatus* and *A. nevadensis* with a/a' and b clearly indicated as dashed vertical lines. When all genetic distances are equal to zero, no step function is shown.

MCMC parameter traceplots showed rapid mixing of chains to the stationary distribution (Figure 3). Further, all $\hat{R}$ and ESS values (both not shown) were close to their recommended cutoffs of one and thousands of samples, respectively, indicating chains are both well-mixed and have converged to the posterior distribution. Bayesian posterior estimates were reported alongside frequentist MLEs, in addition to SEs, posterior SDs, 95% CIs, and 95% CrIs (Table 2).

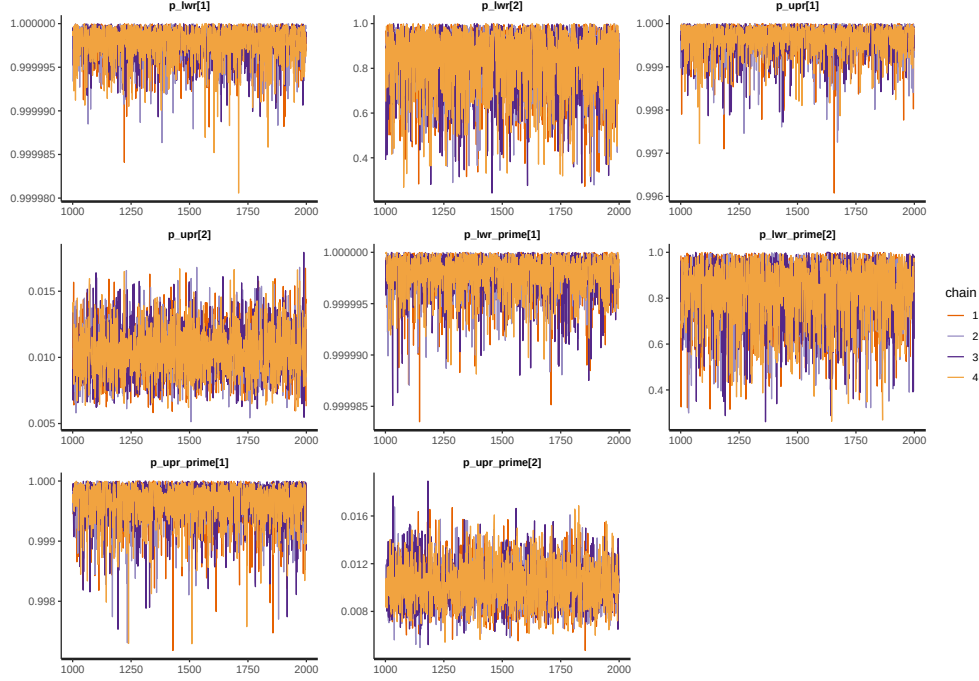


Figure 3: MCMC parameter traceplots applied to *A. bipustulatus* ([1]; $N = 701$) and *A. nevadensis* ([2]; $N = 2$) for CYTB across 1000 post-warmup iterations.

Table 2: Nonparametric frequentist and Bayesian estimates of distance distribution overlap/separation for the DNA barcode gap coalescent model parameters applied to *A. bipustulatus* ($N = 701$) and *A. nevadensis* ($N = 2$) for CYTB, including 95% CIs and CrIs. CrIs are based on 4000 posterior draws. All parameter estimates are reported to three decimal places of precision.

Species	Parameter	MLE (SE, 95% CI)	Bayes Est. (SD; 95% CrI)
<i>A. bipustulatus</i>	p_x/p_{lwr}	1.000 (0.000; 1.000-1.000)	1.000 (0.000; 1.000-1.000)
<i>A. bipustulatus</i>	q_x/p_{upr}	1.000 (0.000; 1.000-1.000)	1.000 (0.000; 0.999-1.000)
<i>A. bipustulatus</i>	p'_x/p'_{lwr}	1.000 (0.000; 1.000-1.000)	1.000 (0.000; 1.000-1.000)
<i>A. bipustulatus</i>	q'_x/p'_{upr}	1.000 (0.000; 1.000-1.000)	1.000 (0.000; 0.999-1.000)
<i>A. nevadensis</i>	p_x/p_{lwr}	1.000 (0.000; 1.000-1.000)	0.835 (0.144; 0.470-0.996)
<i>A. nevadensis</i>	q_x/p_{upr}	0.010 (0.002; 0.006-0.014)	0.010 (0.002; 0.007-0.014)
<i>A. nevadensis</i>	p'_x/p'_{lwr}	1.000 (0.000; 1.000-1.000)	0.834 (0.138; 0.481-0.994)
<i>A. nevadensis</i>	q'_x/p'_{upr}	0.010 (0.070; -0.128-0.148)	0.010 (0.002; 0.007-0.014)

581 CIs were calculated using the usual large sample $(1 - \alpha)100\%$ -level interval estimate given
582 by $\hat{p} \pm z_{1-\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$, where $z_{1-\frac{\alpha}{2}} = 1.960$ is the critical value for 95% confidence
583 (*i.e.*, the 97.5th percent quantile from the standard Normal distribution), and α is the stated
584 significance level (here, 5%). Given a $(1 - \alpha)100\%$ CI, with repeated sampling, on average

$(1 - \alpha)100\%$ of constructed intervals will contain the true parameter of interest, assuming a
correctly-specified statistical model; on the other hand, any given CI will either capture or
exclude the true parameter with 100% certainty. This in stark contrast to a CrI, where the
true parameter is contained within said interval with $(1 - \alpha)100\%$ probability, contingent
on the data generating model being correct. Note, by default Stan computes equal-tailed
(central) CrIs such that there is equal area situated in the left and right tails of the posterior
distribution. For a 95% CrI, this corresponds to the 2.5th and 97.5th percent quantiles.
However, constructed intervals are usually only valid for symmetric or nearly symmetric
distributions. Given the bounded nature of the DNA barcode gap metrics, whose posterior
distributions, as expected, show considerable skewness, an alternative approach to reporting
CrIs, such as Highest Posterior Density (HPD) intervals (Chen and Shao, 1999) or shortest
probability intervals (SPIn) (Liu et al., 2015) is warranted. Although the data for *Agabus*
were not found to be consistent with a value of zero for the metrics (suggesting the existence
of a DNA barcode gap) for either species, this is clearly a case that must be differentiated from
one where some or all of the estimators attain a value of one (indicating no DNA barcode gap
likely exists). As such asymmetric intervals generally attain greater statistical efficiency (in
the form of smaller Mean Squared Error (MSE) or variance) and higher coverage probabilities
than more standard interval estimates, careful in-depth comparison is left for future work.

Findings based on nonparametric MLEs and Bayesian posterior means were quite
comparable with one another and show evidence of complete overlap in intraspecific,
interspecific, and combined interspecific distances for *A. bipustulatus* in both the
 $p/q/p_{\text{lower}}/p_{\text{upper}}$ and $p'/q'/p'_{\text{lower}}/p'_{\text{upper}}$ directions since the metrics and their intervals attain
magnitudes very close or equal to one, with minimal noise (**Table 2**). As a result, this
likely indicates that no DNA barcode gap is present for this species. Such findings are strongly
reinforced by the very tight clustering of posterior draws (**Figure 4**) and associated interval
estimates owing to the large number of specimens sampled for this species.

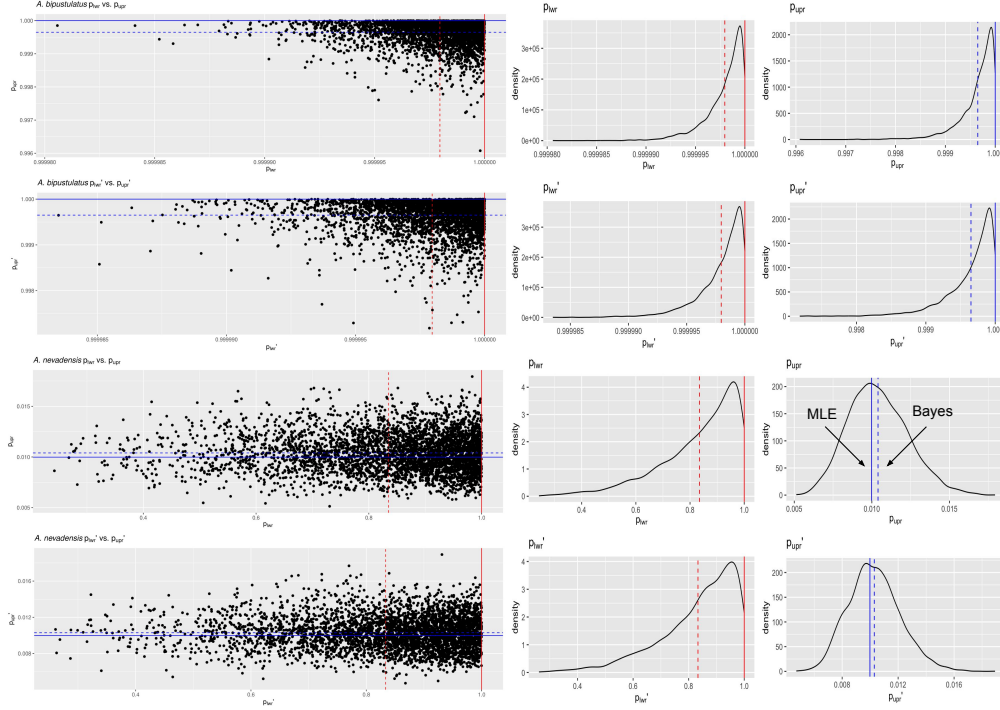


Figure 4: Scatterplots (black solid points) and distributions (black solid lines) depicting the DNA barcode gap metrics for *A. bipustulatus* ($N = 701$) and *A. nevadensis* ($N = 2$) across CYTB based on 4000 Bayesian posterior draws. MLEs and posterior means are displayed as coloured (red/blue) solid and dashed lines for the metrics, respectively.

On the other hand, the situation for *A. nevadensis* is more nuanced, as posterior values are further spread out (**Table 2** and **Figure 4**), suggesting less overall certainty in true parameter values given the low specimen sampling coverage for this taxon. Of note, calculated SEs and SDs are large, with the 95% CIs and 95% CrIs being quite wide in most cases, consistent with much uncertainty regarding the computed frequentist and Bayesian posterior means of the DNA barcode gap metrics. For instance, the Bayesian analysis suggests that the data are consistent with both p_{low} and p'_{low} ranging from approximately 0.500-1.000 due to the large SD of about 0.140. The posterior means associated with these estimates themselves are also far from one. The CrI for p_{upr} and p'_{upr} also spans an order of magnitude, with the lower tail estimate of 0.007 being very close to zero. Further, regarding the frequentist analysis for the same species, the 95% CI for q_x is quite wide, reflecting considerable uncertainty in its true parameter value. Similarly, that for q'_x extends to negative values at the left endpoint,

due to the corresponding SE of 0.070 being too high as a result of the extremely low sample size of $n = 2$ individuals sampled (**Table 2**). Since the 95% CI for q'_x truncated at the lower endpoint includes the value of zero, the null hypothesis for the presence of a DNA barcode gap would not be rejected. Despite this, it is worth noting that truncation is not standard statistical practice and will likely lead to an interval with less than 95% nominal coverage having high relative error. In such cases, more appropriate confidence interval methods like the Wilson score interval, the exact (Clopper-Pearson) interval, or the Agresti-Coull interval should be employed (Newcombe, 1998; Agresti and Coull, 1998). KDEs for *A. bipustulatus* are strongly left (negatively) skewed (**Figure 4**), whereas those for *A. nevadensis* exhibit more symmetry, especially for p_{upr} and p'_{upr} (**Figure 4**). These differences are likely due to the stark contrast in sample sizes for the two examined species. Nevertheless, simulated counts of overlapping specimen records from the posterior predictive distribution (**Table 3**) were found to be very close to observed counts for both species, indicating that the proposed model adequately captures underlying variation.

Table 3: Posterior predictive checks of distance distribution overlap/separation for the DNA barcode gap coalescent model parameters applied to *A. bipustulatus* ($N = 701$) and *A. nevadensis* ($N = 2$) for CYTB. CrIs are based on 4000 posterior draws. All parameter estimates are reported to three decimal places of precision.

Species	Variable	Y	n	Bayes Est. (SD; 95% CrI)
<i>A. bipustulatus</i>	y_{lwr}	491401.000	491401.000	491400.018 (1.378; 491396.000-491401.000)
<i>A. bipustulatus</i>	y_{upr}	2804.000	2804.000	2803.019 (1.433; 2799.000-2804.000)
<i>A. bipustulatus</i>	y'_{lwr}	491401.000	491401.000	491400.008 (1.412; 491396.000-491401.000)
<i>A. bipustulatus</i>	y'_{upr}	2804.000	2804.000	2802.992 (1.429; 2799.000-2804.000)
<i>A. nevadensis</i>	y_{lwr}	4.000	4.000	3.355 (0.888; 1.000-4.000)
<i>A. nevadensis</i>	y_{upr}	28.000	2804.000	29.151 (7.620; 16.000-45.000)
<i>A. nevadensis</i>	y'_{lwr}	4.000	4.000	3.325 (0.884; 1.000-4.000)
<i>A. nevadensis</i>	y'_{upr}	28.000	2804.000	28.942 (7.409; 15.000-45.000)

Obtained results suggest that use of the Beta(1, 1) prior may not be appropriate given a low number of collected individuals for most taxa in DNA barcoding efforts. This suggests that further consideration of more informative beta priors is worthwhile. This is clear for *A. nevadensis*, where use of a stronger prior will likely eliminate the issue regarding accurate estimation of p_{lwr} and p'_{lwr} and stabilize their uncertainty.

4 Conclusion

Herein, the accuracy of the DNA barcode gap was analyzed from a rigorous statistical lens to expedite both the curation and growth of reference sequence libraries, ensuring they are populated with high quality, statistically defensible specimen records fit for purpose to address standing questions in ecology, evolutionary biology, biodiversity forensics, conservation, management, and restoration. To accomplish this, recently proposed, easy to calculate, nonparametric MLEs were formally derived using ECDFs/CECDFs and applied to assess the extent of overlap/separation of distance distributions within and among two species of predatory water beetles in the genus *Agabus* sequenced at CYTB using a Bayesian binomial count model with conjugate beta priors. Findings highlight a high level of parameter uncertainty for *A. nevadensis*, whereas posterior estimates of the DNA barcode gap metrics for *A. bipustulatus* are much more certain. Based on these results, it is imperative that specimen sampling be prioritized to better reflect actual species boundaries. More generally, apart from the metrics being employed to better highlight the importance of within-species genetic diversity versus between-species divergence, it is expected that the approach developed herein will be of broad utility in applied fields, such as DNA-based detection of seafood fraud within global supply chains, and in the determination of species occupancy/detection probabilities at ecological sites of interest using active and passive environmental DNA (eDNA) methods such as metabarcoding.

Since the DNA barcode gap metrics often attain values very close to zero (suggesting no overlap and complete separation of distance distributions) and/or very near one (indicating no separation and complete overlap), in addition to more intermediate values, a noninformative Beta($\frac{1}{2}, \frac{1}{2}$) prior may be more appropriate over complete ignorance imposed by a Beta(1, 1) prior. The former distribution is U-shaped symmetric and places greater probability density at the extremes of the distribution due to its heavier tails, while still allowing for variability in parameter estimates within intermediate values along its domain. This feature is appealing since Phillips et al.'s (2024) metrics often take on values of zero and one. Note that this

669 prior is Jeffreys' prior density (Jeffreys, 1946), which is proportional to the square root of the
 670 Fisher information $\mathcal{I}(\theta)$; that is, $\pi(\theta) \propto \theta^{-\frac{1}{2}}(1 - \theta)^{-\frac{1}{2}}$. Jeffreys' prior has several desirable
 671 statistical properties as a prior: that it is inversely proportional to the standard deviation of
 672 the binomial distribution, and most notably, that it is invariant to model reparameterization
 673 (Gelman et al., 2014). However, this prior can lead to divergent transitions, among other
 674 pathologies, imposed by complex geometry (*i.e.*, curvature) in the posterior space since many
 675 iterative stochastic MCMC sampling algorithms experience difficulties when exploring high
 676 density distribution regions. Thus, remedies to resolve them, such as lowering the step size of
 677 the HMC/NUTS sampler, should be attempted in future work, along with other approaches
 678 such as empirical Bayes estimation to approximate beta prior hyperparameters from observed
 679 data through the MLE or other methods of parameter estimation, such as the method
 680 of moments. Alternatively, hierarchical modelling could be employed to estimate separate
 681 distribution model hyperparameters for each species and/or compute distinct estimates for
 682 the directionality/comparison level of the DNA barcode gap metrics (*i.e.*, lower *vs.* upper,
 683 non-prime *vs.* prime) separately within the genus under study. This would permit greater
 684 flexibility through incorporating more fine-grained structure seen in the data. However, low
 685 taxon sample sizes may preclude valid inferences to be reasonably ascertained due to the
 686 large number additional parameters which would be introduced through the specification
 687 of the hyperprior distributions. Methods outlined in Gelman et al. (2014, 2020), such as
 688 dealing with non-exchangeability of observations and alternate model parameterizations like
 689 the logit, may prove useful in this regard. Perhaps, the best course of action though is
 690 the elicitation of priors from domain experts. Even though more work remains, it is clear
 691 that both frequentist and Bayesian inference hold much promise for the future of molecular
 692 biodiversity science.

Supplementary Information

None declared

Data Availability Statement

Raw data, R, and Stan code can be accessed via Dryad at:

<http://datadryad.org/stash/share/>

RZIfMixcEODe0RWP7eyXWQewSVbqEIA9UTrH3ZVKyn4.

A GitHub repository can be found at:

<https://github.com/jphill01/Bayesian-DNA-Barcode-Gap-Coalescent>.

Acknowledgements

We wish to recognise the valuable comments and discussions of Daniel (Dan) Gillis, Robert (Bob) Hanner, Kate Lindsay, Robert (Rob) Young, the handling editor, and XXX anonymous reviewers.

We acknowledge that the University of Guelph resides on the ancestral lands of the Attawandaron people and the treaty lands and territory of the Mississaugas of the Credit. We recognize the significance of the Dish with One Spoon Covenant to this land and offer our respect to our Anishinaabe, Haudenosaunee and Métis neighbours as we strive to strengthen our relationships with them.

Funding

None declared.

Conflict of Interest

None declared.

Author Contributions

JDP wrote the manuscript, wrote R and Stan code, as well as analyzed and interpreted all model results.

References

Agresti, A. and B. A. Coull

1998. Approximate is better than ‘exact’ for interval estimation of binomial proportions.

The American Statistician, 52(2):119–126.

Ahrens, D., F. Fujisawa, H.-J. Krammer, J. Eberle, S. Fabrizi, and A. Vogler

2016. Rarity and incomplete sampling in DNA-based species delimitation. *Systematic*

Biology, 65(3):478–494.

Avise, J., J. Arnold, R. Ball, Jr., E. Bermingham, T. Lamb, J. Neigel, C. Reeb, and
N. Saunders

1987. Intraspecific phylogeography: The mitochondrial DNA bridge between population

genetics and systematics. *Annu. Rev. Ecol. Syst.*, 18:489–522.

Bartlett, S. and W. Davidson

1992. FINS (forensically informative nucleotide sequencing): A procedure for identifying

the animal origin of biological specimens. *BioTechniques*, 12(3):408–411.

Bergsten, J., D. Bilton, T. Fujisawa, M. Elliott, M. Monaghan, M. Balke, L. Hendrich,

- J. Geijer, J. Herrmann, G. Foster, I. Ribera, A. Nilsson, T. Barraclough, and A. Vogler
2012. The effect of geographical scale of sampling on DNA barcoding. *Systematic Biology*,
61(5):851–869.
- Bouckaert, R., T. G. Vaughan, J. Barido-Sottani, S. Duchêne, M. Fourment,
A. Gavryushkina, J. Heled, G. Jones, D. Kühnert, N. De Maio, M. Matschiner, F. K.
Mendes, N. F. Müller, H. A. Ogilvie, L. du Plessis, A. Poppinga, A. Rambaut, D. Rasmussen,
I. Siveroni, M. A. Suchard, C.-H. Wu, D. Xie, C. Zhang, T. Stadler, and A. J. Drummond
2019. BEAST 2.5: An advanced software platform for Bayesian evolutionary analysis.
PLOS Computational Biology, 15(4):1–28.
- Bucklin, A., D. Steinke, and L. Blanco-Bercial
2011. DNA barcoding of marine Metazoa. *Annual Review of Marine Science*, 3(Volume 3,
2011):471–508.
- Camargo, A., M. Morando, L. J. Avila, and J. W. Sites Jr.
2012. Species delimitation with abc and other coalescent-based methods: A test of accuracy
with simulations and an empirical example with lizards of the *Liolaemus darwini* complex
(squamata: Liolaemidae). *Evolution*, 66(9):2834–2849.
- Čandek, K. and M. Kuntner
2015. DNA barcoding gap: Reliable species identification over morphological and
geographical scales. *Molecular Ecology Resources*, 15(2):268–277.
- Carpenter, B., A. Gelman, M. Hoffman, D. Lee, B. Goodrich, M. Betancourt, M. Brubaker,
J. Guo, P. Li, and A. Riddell
2017. Stan: A probabilistic programming language. *Journal of Statistical Software*, 76:1.
- Carstens, B. C., T. A. Pelletier, N. M. Reid, and J. D. Satler
2013. How to fail at species delimitation. *Molecular Ecology*, 22(17):4369–4383.

756 Chen, M.-H. and Q.-M. Shao
757 1999. Monte Carlo estimation of Bayesian credible and HPD intervals. *Journal of*
758 *Computational and Graphical Statistics*, 8(1):69–92.

759 Collins, R. A. and R. H. Cruickshank
760 2013. The seven deadly sins of DNA barcoding. *Molecular Ecology Resources*,
761 13(6):969–975.

762 Collins, R. A. and R. H. Cruickshank
763 2014. Known knowns, known unknowns, unknown unknowns and unknown knowns in
764 DNA barcoding: A comment on Dowton et al. *Systematic Biology*, 63(6):1005–1009.

765 Dayrat, B.
766 2005. Towards integrative taxonomy. *Biological Journal of the Linnean Society*,
767 85(3):407–417.

768 de Queiroz, K.
769 2007. Species concepts and species delimitation. *Systematic Biology*, 56(6):879–886.

770 De Sanctis, B., D. Money, M. Pedersen, and R. Durbin
771 2022. A theoretical analysis of taxonomic binning accuracy. *Molecular Ecology Resources*,
772 22(7):2208–2219.

773 Degnan, J. H. and N. A. Rosenberg
774 2009. Gene tree discordance, phylogenetic inference and the multispecies coalescent. *Trends*
775 *in Ecology & Evolution*, 24(6):332–340.

776 Dellicour, S. and J.-F. Flot
777 2015. Delimiting species-poor data sets using single molecular markers: A study of barcode
778 gaps, haplowebs and GMYC. *Systematic Biology*, 64(6):900–908.

779 Dellicour, S. and J.-F. Flot
780 2018. The hitchhiker’s guide to single-locus species delimitation. *Molecular Ecology*
781 *Resources*, 18(6):1234–1246.

782 Dempster, A. P., N. M. Laird, and D. B. Rubin
783 1977. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the*
784 *Royal Statistical Society: Series B (Methodological)*, 39(1):1–22.

785 Dowton, M., K. Meiklejohn, S. L. Cameron, and J. Wallman
786 2014. A preliminary framework for DNA barcoding, incorporating the multispecies
787 coalescent. *Systematic Biology*, 63(4):639–644.

788 Efron, B.
789 1979. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*,
790 7(1):1–26.

791 Efron, B., E. Halloran, and S. Holmes
792 1996. Bootstrap confidence levels for phylogenetic trees. *Proceedings of the National*
793 *Academy of Sciences of the United States of America*, 93(23):13429–13434.

794 Efron, B. and R. J. Tibshirani
795 1993. *An Introduction to the Bootstrap*. Chapman & Hall.

796 Flouri, T., J. Huang, X. Jiao, P. Kapli, B. Rannala, and Z. Yang
797 2022. Bayesian phylogenetic inference using relaxed-clocks and the multispecies coalescent.
798 *Molecular Biology and Evolution*, 39(msac161).

799 Flouri, T., X. Jiao, J. Huang, B. Rannala, and Z. Yang
800 2023. Efficient Bayesian inference under the multispecies coalescent with migration.
801 *Proceedings of the National Academy of Sciences of the United States of America*,
802 120(e2310708120).

- Flouri, T., X. Jiao, B. Rannala, and Z. Yang
2018. Species tree inference with BPP using genomic sequences and the multispecies
coalescent. *Molecular Biology and Evolution*, 35(10):2585–2593.
- Flouri, T., X. Jiao, B. Rannala, and Z. Yang
2020. A Bayesian implementation of the multispecies coalescent model with introgression
for phylogenomic analysis. *Molecular Biology and Evolution*, 37(4):1211–1223.
- Fonseca, E. M., D. J. Duckett, and B. C. Carstens
2021. P2C2M.GMYC: An R package for assessing the utility of the Generalized Mixed
Yule Coalescent model. *Methods in Ecology and Evolution*, 12(3):487–493.
- Fujisawa, T. and T. Barraclough
2013. Delimiting species using single-locus data and the generalized mixed yule coalescent
approach: A revised method and evaluation on simulated data sets. *Systematic Biology*,
62(5):707–724.
- Fujita, M. K., A. D. Leaché, F. T. Burbrink, J. A. McGuire, and C. Moritz
2012. Coalescent-based species delimitation in an integrative taxonomy. *Trends in Ecology
& Evolution*, 27(9):480–488.
- Gelman, A., J. Carlin, H. Stern, D. Duncan, A. Vehtari, and D. Rubin
2014. *Bayesian Data Analysis*, third edition. Chapman and Hall/CRC.
- Gelman, A. and D. Rubin
1992. Inference from iterative simulation using multiple sequences. *Statistical Science*,
7(4):457–472.
- Gelman, A., A. Vehtari, D. Simpson, C. Margossian, B. Carpenter, Y. Yao, L. Kennedy,
J. Gabry, P.-C. Bürkner, and M. Modrák
2020. Bayesian workflow.

Hart, M. and J. Sunday

2007. Things fall apart: Biological species form unconnected parsimony networks. *Biology Letters*, 3(5):509–512.

Hebert, P., A. Cywinska, S. Ball, and J. deWaard

2003a. Biological identifications through DNA barcodes. *Proceedings of the Royal Society of London B: Biological Sciences*, 270(1512):313–321.

Hebert, P., S. Ratnasingham, and J. de Waard

2003b. Barcoding animal life: Cytochrome c oxidase subunit 1 divergences among closely related species. *Proceedings of the Royal Society of London B: Biological Sciences*, 270(Suppl 1):S96–S99.

Hebert, P. D., M. Y. Stoeckle, T. S. Zemlak, and C. M. Francis

2004. Identification of birds through DNA barcodes. *PLoS Biol*, 2(10):e312.

Hickerson, M. J., C. P. Meyer, and C. Moritz

2006. DNA barcoding will often fail to discover new animal species over broad parameter space. *Systematic Biology*, 55(5):729–739.

Ho, S. and S. Duchêne

2014. Molecular-clock methods for estimating evolutionary rates and timescales. *Molecular Ecology*, 23(24):5947–5965.

Hoffman, M. and A. Gelman

2014. The No-U-Turn Sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15:1593–1623.

Hubert, N. and R. Hanner

2015. DNA barcoding, species delineation and taxonomy: A historical perspective. *DNA Barcodes*, 3:44–58.

Hubert, N., J. Phillips, and R. Hanner

2024. Delimiting species with single-locus DNA sequences. In *DNA Barcoding: Methods and Protocols*, R. DeSalle, ed., volume 2744 of *Methods in Molecular Biology*. New York, NY: Humana.

Jackson, N. D., B. C. Carstens, A. E. Morales, and B. C. O’Meara

2017a. Species delimitation with gene flow. *Systematic Biology*, 66(5):799–812.

Jackson, N. D., A. E. Morales, B. C. Carstens, and B. C. O’Meara

2017b. PHRAPL: Phylogeographic inference using approximate likelihoods. *Systematic Biology*, 66(6):1045–1053.

Jeffreys, H.

1946. An invariant form for the prior probability in estimation problems. *Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences*, 186(1007):453–461.

Jukes, T. and C. Cantor

1969. Evolution of protein molecules. In *Mammalian Protein Metabolism*, H. N. Munro, ed., Pp. 21–132. New York: Academic Press.

Kapli, P., S. Lutteropp, J. Zhang, K. Kobert, P. Pavlidis, A. Stamatakis, and T. Flouri

2017. Multi-rate poisson tree processes for single-locus species delimitation under maximum likelihood and markov chain monte carlo. *Bioinformatics*, 33(11):1630–1638.

Kimura, M.

1980. A simple method for estimating evolutionary rates of base substitutions through comparative studies of nucleotide sequences. *Journal of Molecular Evolution*, 16(1):111–120.

Kingman, J.

1982a. The coalescent. *Stochastic Processes and Their Applications*, 13:235–248.

876 Kingman, J.
877 1982b. On the genealogy of large populations. *Journal of Applied Probability*, 19(A):27–43.

878 Knowles, L. L.
879 2004. The burgeoning field of statistical phylogeography. *Journal of Evolutionary Biology*,
880 17(1):1–10.

881 Knowles, L. L.
882 2009. Statistical phylogeography. *Annual Review of Ecology, Evolution, and Systematics*,
883 40(40):593–612.

884 Knowles, L. L. and W. P. Maddison
885 2002. Statistical phylogeography. *Molecular Ecology*, 11(12):2623–2635.

886 Kullback, S. and R. Leibler
887 1951. On information and sufficiency. *Annals of Mathematical Statistics*, 22(1):79–86.

888 Leaché, A., M. Fujita, V. Minin, and R. Bouckaert
889 2014. Species delimitation using genome-wide SNP data. *Systematic Biology*,
890 63(4):534–542.

891 Liu, Y., A. Gelman, and T. Zheng
892 2015. Simulation-efficient shortest probability intervals. *Statistical Computing*, 25:809–819.

893 Mather, N., S. Traves, and S. Ho
894 2019. A practical introduction to sequentially Markovian coalescent methods for estimating
895 demographic history from genomic data. *Ecology and Evolution*, 10(1):579–589.

896 Matz, M. V. and R. Nielsen
897 2005. A likelihood ratio test for species membership based on dna sequence
898 data. *Philosophical Transactions of the Royal Society B: Biological Sciences*,
899 360(1459):1969–1974.

- 900 Mayr, E.
901 1942. *Systematics and the Origin of Species from the Viewpoint of a Zoologist*. New York:
902 Columbia University Press.
- 903 Meier, R., G. Zhang, and F. Ali
904 2008. The use of mean instead of smallest interspecific distances exaggerates the size of
905 the “barcoding gap” and leads to misidentification. *Systematic Biology*, 57(5):809–813.
- 906 Meyer, C. and G. Paulay
907 2005. DNA barcoding: Error rates based on comprehensive sampling. *PLOS Biology*,
908 3(12):e422.
- 909 Monaghan, M., R. Wild, M. Elliot, T. Fujisawa, M. Balke, D. Inward, D. Lees,
910 R. Ranaivosolo, P. Eggleton, T. Barraclough, and A. Vogler
911 2009. Accelerated species inventory on Madagascar using coalescent-based models of
912 species delineation. *Systematic Biology*, 58:298–311.
- 913 Mutanen, M., S. M. Kivelä, R. A. Vos, C. Doorenweerd, S. Ratnasingham, A. Hausmann,
914 P. Huemer, V. Dincă, E. J. van Nieukerken, C. Lopez-Vaamonde, R. Vila, L. Aarvik,
915 T. Decaëns, K. A. Efetov, P. D. N. Hebert, A. Johnsen, O. Karsholt, M. Pentinsaari,
916 R. Rougerie, A. Segerer, G. Tarmann, R. Zahiri, and H. C. J. Godfray
917 2016. Species-level para- and polyphyly in DNA barcode gene trees: Strong operational
918 bias in european lepidoptera. *Systematic Biology*, 65(6):1024–1040.
- 919 Neal, R.
920 2011. MCMC using Hamiltonian dynamics. In *Handbook of Markov Chain Monte Carlo*.
921 Chapman & Hall, CRC Press.
- 922 Newcombe, R. G.
923 1998. Two-sided confidence intervals for the single proportion: comparison of seven
924 methods. *Statistics in Medicine*, 17(8):857–872.

925 Nielsen, R. and M. Matz
926 2006. Statistical approaches for dna barcoding. *Systematic Biology*, 55(1):162–169.

927 Pante, E., N. Puillandre, A. Viricel, S. Arnaud-Haond, D. Aurelle, M. Castelin, A. Chenuil,
928 C. Destombe, D. Forcioli, M. Valero, F. Viard, and S. Samadi
929 2015. Species are hypotheses: Avoid connectivity assessments based on pillars of sand.
930 *Molecular Ecology*, 24(3):525–544.

931 Paradis, E., J. Claude, and K. Strimmer
932 2004. APE: Analyses of Phylogenetics and Evolution in R language. *Bioinformatics*,
933 20(2):289–290.

934 Pentinsaari, M., H. Salmela, M. Mutanen, and T. Roslin
935 2016. Molecular evolution of a widely-adopted taxonomic marker (COI) across the animal
936 tree of life. *Scientific Reports*, 6:35275.

937 Phillips, J., T. Athey, P. McNicholas, and R. Hanner
938 2023. VLF: An R package for the analysis of very low frequency variants in DNA sequences.
939 *Biodiversity Data Journal*, 11:e96480.

940 Phillips, J., D. Gillis, and R. Hanner
941 2019. Incomplete estimates of genetic diversity within species: Implications for DNA
942 barcoding. *Ecology and Evolution*, 9(5):2996–3010.

943 Phillips, J., D. Gillis, and R. Hanner
944 2020. HACSim: An R package to estimate intraspecific sample sizes for genetic diversity
945 assessment using haplotype accumulation curves. *PeerJ Computer Science*.

946 Phillips, J., D. Gillis, and R. Hanner
947 2022. Lack of statistical rigor in DNA barcoding likely invalidates the presence of a true
948 species’ barcode gap. *Frontiers in Ecology and Evolution*, 10:859099.

Phillips, J., C. Griswold, R. Young, N. Hubert, and H. Hanner

2024. *A Measure of the DNA Barcode Gap for Applied and Basic Research*, Pp. 375–390.
New York, NY: Springer US.

Phillips, J., R. Gwiazdowski, D. Ashlock, and R. Hanner

2015. An exploration of sufficient sampling effort to describe intraspecific DNA barcode
haplotype diversity: examples from the ray-finned fishes (Chordata: Actinopterygii). *DNA
Barcodes*, 3:66–73.

Pons, J., T. G. Barraclough, J. Gomez-Zurita, A. Cardoso, D. P. Duran, S. Hazell,
S. Kamoun, W. D. Sumlin, and A. P. Vogler

2006. Sequence-based species delimitation for the DNA taxonomy of undescribed insects.
Systematic Biology, 55(4):595–609.

Puillandre, N., S. Brouillet, and G. Achaz

2021. ASAP: Assemble Species by Automatic Partitioning. *Molecular Ecology Resources*,
21(2):609–620.

Puillandre, N., A. Lambert, S. Brouillet, and G. Achaz

2011. ABGD, Automatic Barcode Gap Discovery for primary species delimitation.
Molecular Ecology, 21(8):1864–1877.

R Core Team

2024. *R: A Language and Environment for Statistical Computing*. R Foundation for
Statistical Computing, Vienna, Austria.

Rannala, B.

2015. The art and science of species delimitation. *Current Zoology*, 61(5):846–853.

Rannala, B. and Z. Yang

2003. Bayes estimation of species divergence times and ancestral population sizes using
DNA sequences from multiple loci. *Genetics*, 164:1645–1656.

974 Rannala, B. and Z. Yang
975 2017. Efficient Bayesian species tree inference under the multispecies coalescent. *Systematic*
976 *Biology*, 66(5):823–842.

977 Ratnasingham, S. and P. Hebert
978 2007. BOLD: The Barcode of Life Data System (<http://www.barcodinglife.org>). *Molecular*
979 *Ecology Notes*, 7(3):355–364.

980 Ratnasingham, S. and P. D. N. Hebert
981 2013. A DNA-based registry for all animal species: The barcode index number (BIN)
982 system. *PLoS One*, 8(7):e66213.

983 Rosenberg, N. and M. Nordborg
984 2002. Genealogical trees, coalescent theory and the analysis of genetic polymorphisms.
985 *Nature Reviews Genetics*, 3(5):380–390.

986 Stan Development Team
987 2023. RStan: The R interface to Stan. R package version 2.32.6.

988 Stephens, M.
989 2000. Dealing with label switching in mixture models. *Journal of the Royal Statistical*
990 *Society: Series B (Statistical Methodology)*, 62(4):795–809.

991 Stoeckle, M. and D. Thaler
992 2014. DNA barcoding works in practice but not in (neutral) theory. *PLoS One*,
993 9(7):e100755.

994 Talavera, G., V. Dincă, and R. Vila
995 2013. Factors affecting species delimitations with the GMYC model: insights from a
996 butterfly survey. *Methods in Ecology and Evolution*, 4(12):1101–1110.

997 Tang, C. Q., A. M. Humphreys, D. Fontaneto, and T. G. Barraclough
 998 2014. Effects of phylogenetic reconstruction method on the robustness of species
 999 delimitation using single-locus data. *Methods in Ecology and Evolution*, 5(10):1086–1094.

1000 Templeton, A.
 1001 1998. Nested clade analyses of phylogeographic data: testing hypotheses about gene flow
 1002 and population history. *Molecular Ecology*, 7(4):381–397.

1003 Templeton, A., K. Crandall, and C. Sing
 1004 1992. A cladistic analysis of phenotypic associations with haplotypes inferred from
 1005 restriction endonuclease mapping and DNA sequence data. III. Cladogram estimation.
 1006 *Genetics*, 132(2):619–633.

1007 Vehtari, A., A. Gelman, D. Simpson, B. Carpenter, and P.-C. Bürkner
 1008 2021. Rank-normalization, folding, and localization: An improved $\hat{R}$ for assessing
 1009 convergence of MCMC (with discussion). *Bayesian Analysis*, 16(2):667–718.

1010 Wells, T., T. Carruthers, P. Muñoz-Rodríguez, A. Sumadijaya, J. R. I. Wood, and R. W.
 1011 Scotland
 1012 2022. Species as a heuristic: Reconciling theory and practice. *Systematic Biology*,
 1013 71(5):1233–1243.

1014 Wickham, H.
 1015 2016. *ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag New York.

1016 Wiemers, M. and K. Fiedler
 1017 2007. Does the DNA barcoding gap exist? - a case study in blue butterflies (Lepidoptera:
 1018 Lycaenidae). *Frontiers in Zoology*, 4(8).

1019 Yang, Z. and B. Rannala
 1020 2010. Bayesian species delimitation using multilocus sequence data. *Proceedings of the*
 1021 *National Academy of Sciences*, 107:9264–9269.

1022 Yang, Z. and B. Rannala
 1023 2014. Unguided species delimitation using DNA sequence data from multiple loci.
 1024 *Molecular Biology and Evolution*, 31(12):3125–3135.

1025 Yang, Z. and B. Rannala
 1026 2017. Bayesian species identification under the multispecies coalescent provides significant
 1027 improvements to DNA barcoding analyses. *Molecular Ecology*, 26:3028–3036.

1028 Young, R., R. Gill, D. Gillis, and R. Hanner
 1029 2021. Molecular Acquisition, Cleaning and Evaluation in R (MACER) - A tool to assemble
 1030 molecular marker datasets from BOLD and GenBank. *Biodiversity Data Journal*, 9:e71378.

1031 Zhang, J., P. Kapli, P. Pavlidis, and A. Stamatakis
 1032 2013. A general species delimitation method with applications to phylogenetic placements.
 1033 *Bioinformatics*, 29(22):2869–2876.